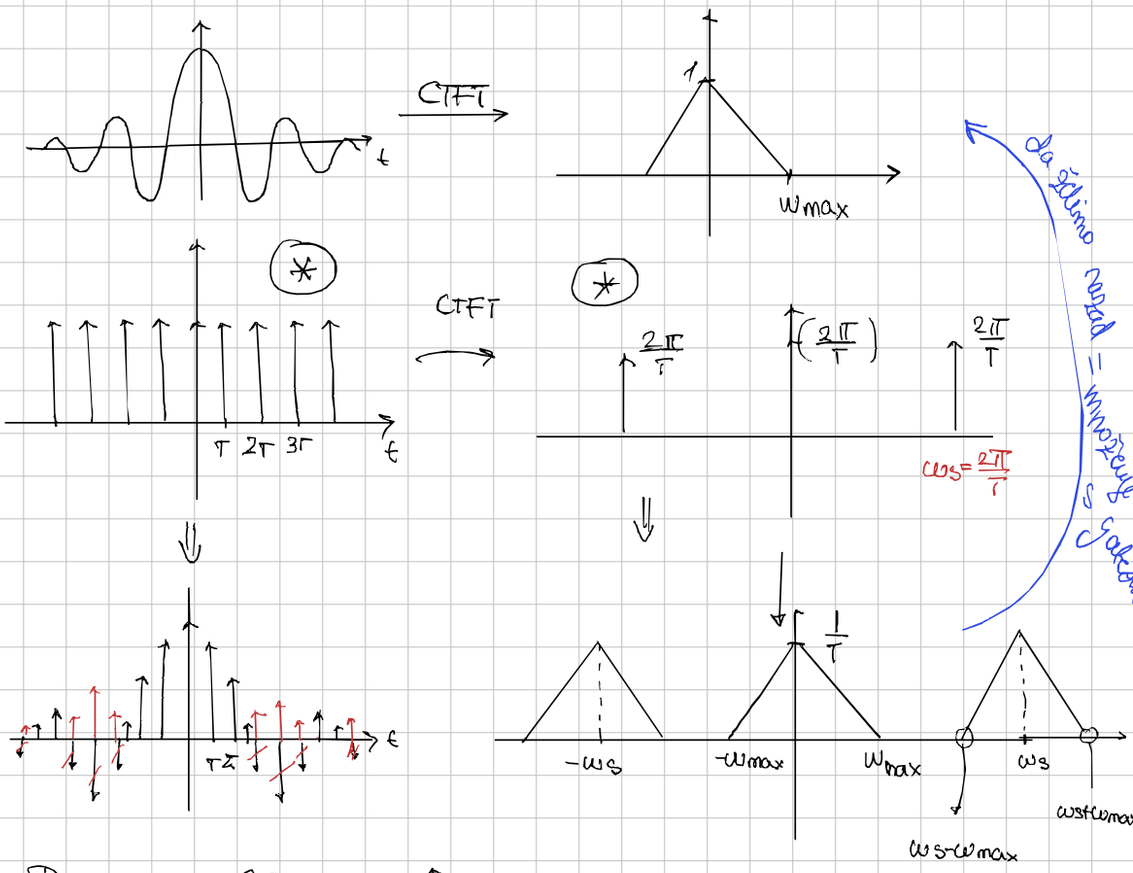
Ponavljanje onog što je Zvonko pokušao ispredavati:



Dirac je u CT → ima ↑

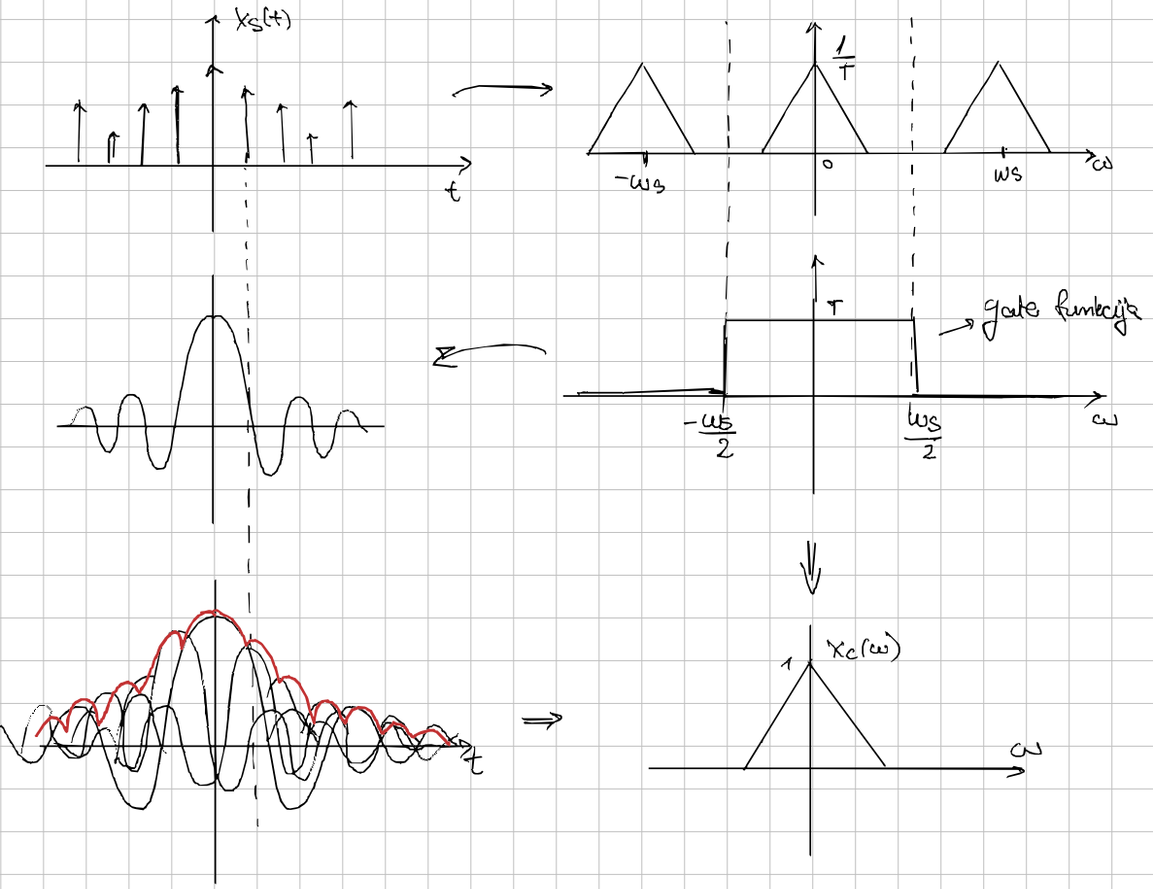
Kronicker je u DT → ima 1

$\omega_{max} < \omega_s - \omega_{max}$

$\boxed{\omega_s > 2\omega_{max}}$ da ne dođe do ALIASINGA (preklapanja)

→ to pazi kad vraćanje

Vraćanje:



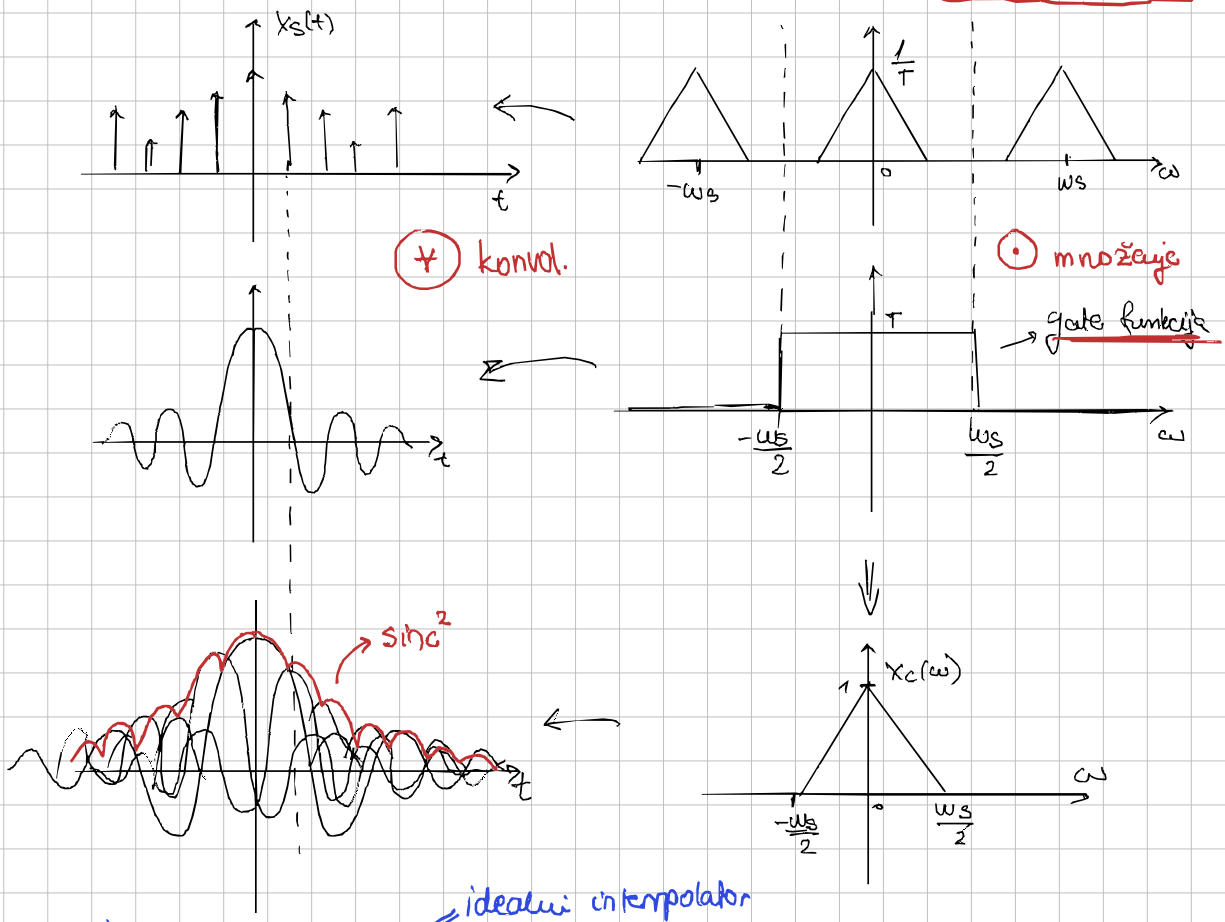
nulti red → stepenice

prvi red → kosi crte

idealno → konvolucija sa sinkom

REKONSTRUKCIJA SIGNALA

sampling → množenje → kopiranje spekt.
rekonstrukcija → konvolucija → filtriranje spektra



idealni interpolator

↳ idealna rekonstrukcija : sinc nije ograničen
= u realnom svijetu ne možemo raditi s takvim signalima

1) imamo OG analogni signal (npr neki smooth val)

2) Uzorkujemo ga → (konvoluc u vremenskoj = množenje u frekv. i obrnuto)

* kada radimo rekonstrukciju ⇒ konvolucija u vremenskoj
* obrnuto (sampling uzorkovanje) ⇒ množenje u vremenskoj

3) dobivamo diskretne vrijednosti u vremenu (još uvijek „analogno" jer impulsi nisu diskretni sami po sebi)

4) Da bismo REKONSTRUIRALI : konvolucija sa sinc

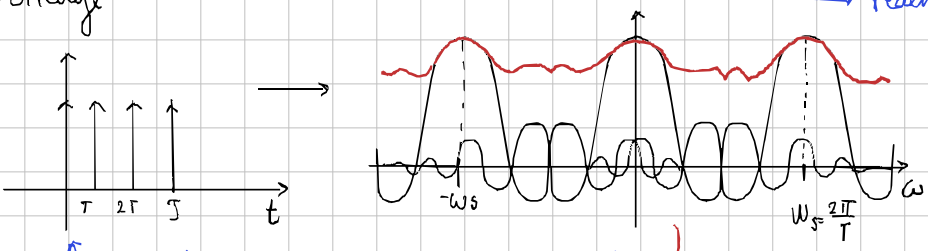
(rekonstrukc.)
interpolator nultog reda = zadržava vrijednost dok ne dođe do sljedećeg trenutka (stepenice)

interpolator prvog reda = pravci

▶ Antialiasing filtri

→ idealni aliasing

očitanje



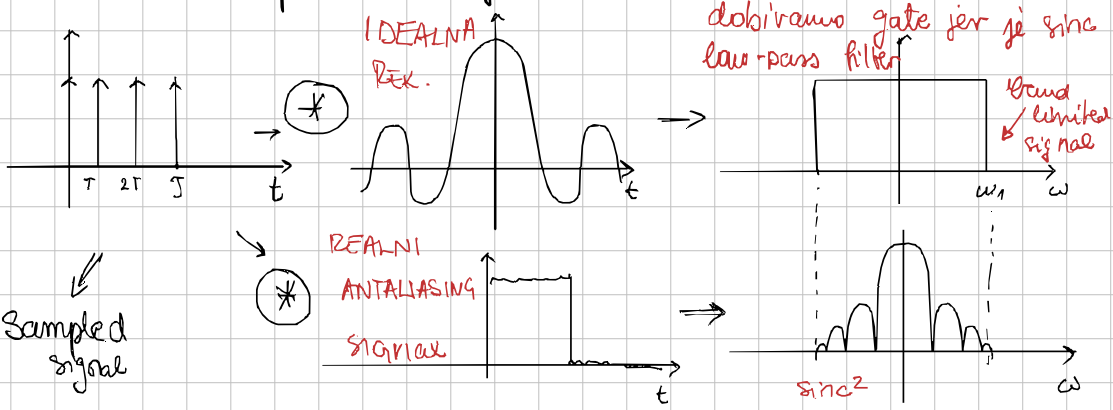
→ realni aliasing filter
- ne koristi se pravi sinc nego nešto što liči na njega

množenje s Diracom pomiče F(ω)

pojas propuštanja i gušenja u ω domeni

kako prepoznati je li sinc ili pravokutni ili nešto treće

→ nastavak: probamo sljedeće



* Dirac nije prava funkcija već generalizirana → u konvoluciji neutralni el. samo pomiče

Sampled signal = niz Dirac impulsa

množenje
- uzorkovana verzija

konvolucija
- sa sincom = rekonstrukcija (prvi red)

idealni rekonstrukcijski filter u frekvenciji = pravokutnik
→ njegov impulsni odziv u vremenu = sinc

realni filter u frekvenciji ≠ savršeni pravokutnik
→ impulsni odziv = nešto brže trajanje, tipa sinc²

Očitavanje u frekvencijskoj domeni

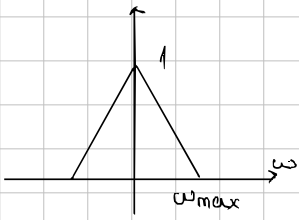
Vremenska domena:

$\sin k^2$

CTFT →

Frekvencijska domena:

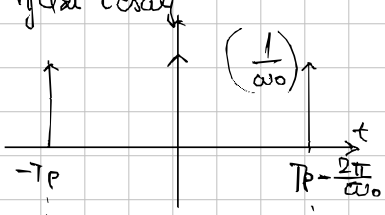
spektar:



konvolucija (*)

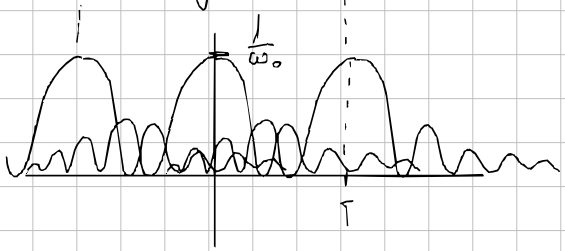
MNOŽENJE (·) s. sa češljem (da dobijemo očitani spektar)

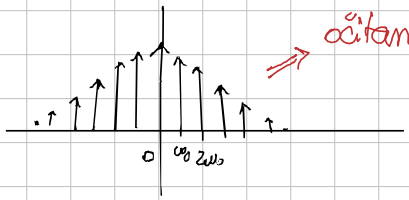
rijetki češalj



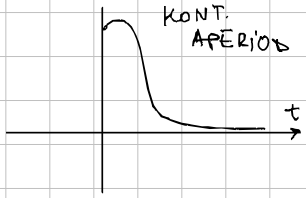
CTFT →

(gusti)

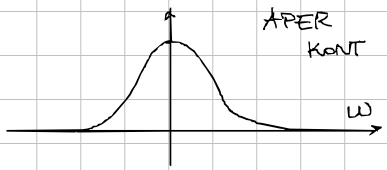




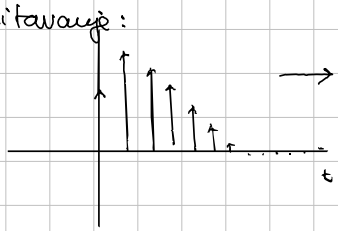
očitani spektar



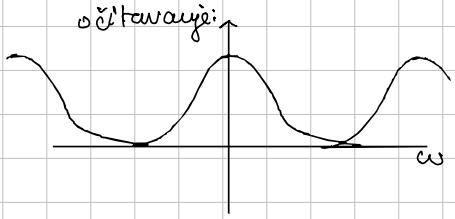
CTFT →

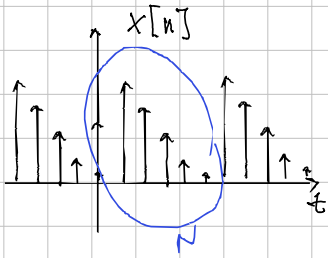


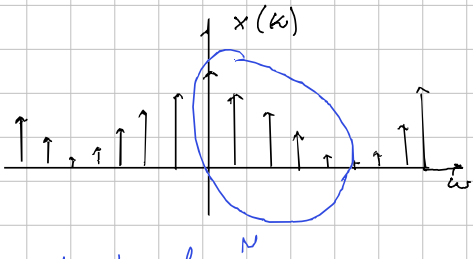
očitavanje:



očitavanje:







ne postoji transf. za nve za koje smo radili

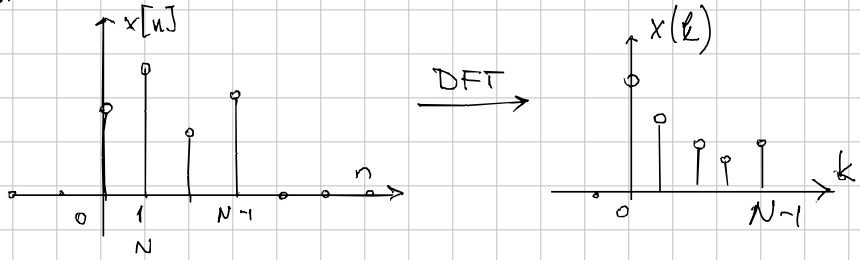
DFT - diskretna Fourierova transformacija

DFT Diskr. aperiodičnu pretvara u diskr. aperiod.

IDFT

$$X(k) = \sum_{n=0}^{N-1} x[n] \cdot e^{-j\frac{2\pi}{N}kn}$$

$$x[n] = \frac{1}{n} \sum_{k=0}^{N-1} X(k)\, e^{j\frac{2\pi}{N}kn}$$



Primjer: $x[n] = \{2, 1, 3, 1\}$, $N = 4$

$$X(k) = \sum_{n=0}^{3} x[n] \cdot e^{-j\frac{2\pi}{N}kn} = 2 \cdot e^{-j\frac{2\pi}{4}\cdot k\cdot 0} + e^{-j\frac{2\pi}{4}\cdot k\cdot 1} + 3 \cdot e^{-j\frac{2\pi}{4}\cdot k\cdot 2} + e^{-j\frac{2\pi}{4}\cdot k\cdot 3}$$

| uvrštavamo $k = 0, 1, 2, 3$ | $= 2 + \cos\left(\frac{\pi}{2}k\right) - j\sin\left(\frac{\pi}{2}k\right) + 3\cos(\pi k) - 3j\sin(\pi k) + \cos\left(\frac{3\pi}{2}k\right) - j\sin\left(\frac{3\pi}{2}k\right)$

$\Downarrow$

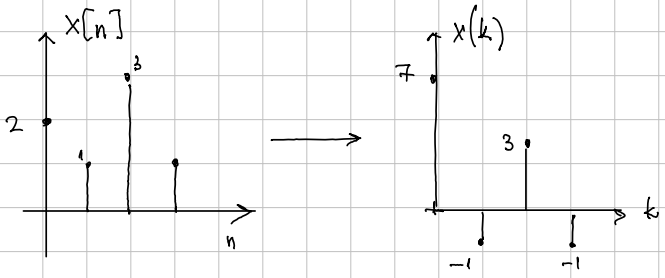
$$X(0) = 2 + 1 - \cancel{j0} + 3 - \cancel{3j0} + 1 - \cancel{j0} = 7$$

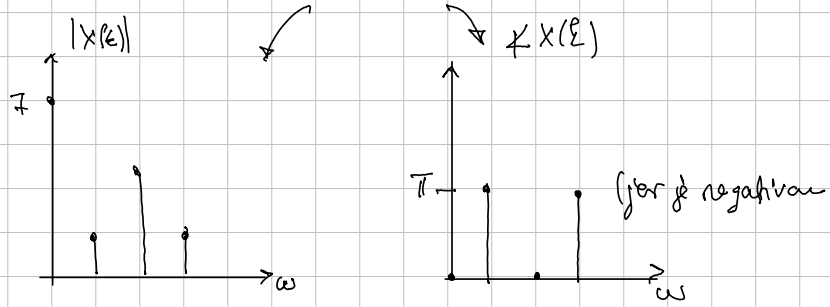
$$X(1) = 2 + 0 - \cancel{j} - 3 - 0 + 0 + \cancel{j} = -1$$

$$X(2) = 2 - 1 - 0 + 3 - 0 - 1 - 0 = 3$$

$$X(3) = 2 + 0 + j - 3 - 0 + 0 - j = -1$$

$$\Rightarrow X(k) = \{7, -1, 3, -1\}$$





Primjer IDFT: Zadan spektar $X[k] = \{7, -1, 3, -1\}_{N=4}$

$$x[n] = \frac{1}{4}\sum_{k=0}^{3} X[k]\cdot e^{j\frac{2\pi}{N}kn} = \frac{1}{4}\left(7e^{j\frac{2\pi}{4}\cdot 0\cdot n} + (-1)\cdot e^{j\frac{2\pi}{4}\cdot 1\cdot n} + 3\cdot e^{j\frac{2\pi}{4}\cdot 2\cdot n} + (-1)\cdot e^{j\frac{2\pi}{4}\cdot 3\cdot n}\right)$$

$$= \frac{1}{4}\left(7 - e^{j\frac{\pi}{2}n} + 3e^{j\pi n} - e^{j\frac{3\pi}{2}n}\right)$$

$$= \frac{1}{4}\left[7 - \cos\left(\frac{\pi}{2}n\right) - j\sin\left(\frac{\pi}{2}n\right) + 3\cos(\pi n) + 3j\sin(\pi n) - \cos\left(\frac{3\pi}{2}n\right) - j\sin\left(\frac{3\pi}{2}n\right)\right]$$

$$\rightarrow x[0] = [7 - 1 + 3 - 1) = 2$$

$$x[1] = [7 - 0 - j - 3 + j] = 1$$

$$x[2] = 3$$

$$x[3] = 1$$

$$x[n] = \{2, 1, 3, 1\}$$